ASOF Join

Team Members

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Overview

- ASOF join matches each left row to the nearest right row by time (e.g. each trade to the latest quote at or before it), optionally per key
- Supported by DuckDB, ClickHouse, Snowflake, kdb+ and QuestDB; missing in PostgreSQL
- Current workaround is a hand-written LEFT JOIN LATERAL ... LIMIT 1: verbose, and slow without an index

Project Goals

1. Implement support for `ASOF JOIN ... MATCH_CONDITION (...) ON ...` syntax with `>=`, `>`, `<=`, `<`
2. Left-outer semantics, correct NULL handling, clear errors
3. MVP by rewriting to a LATERAL join
4. Native merge-based ASOF join in the planner and executor
5. Benchmark native vs. MVP

Implementation Plan

Phase 1: MVP by query rewrite

- Teach PostgreSQL to recognize the new ASOF JOIN syntax
- Translate each ASOF join into an equivalent LATERAL join that picks the single closest matching row, so the existing engine runs it unchanged
- Reject invalid queries with clear error messages, such as a missing or malformed time condition

Phase 2: Native ASOF join

- Make ASOF a real join type that the query planner understands and can pick a strategy for
- Add a new execution method that sorts both tables once and walks through them together, instead of searching for a match row by row
- Show the new join clearly in query plans (EXPLAIN)